# Deepak Garg

+91 7838630502 gargdeepak114@gmail.com linkedin.com/in/deepak-garg-in github.com/mrgarg-g1

## Professional Summary

Data Scientist with 3+ years of experience building and deploying machine learning solutions in production environments. Currently working on large-scale FMCG use cases, focusing on solving high-impact business problems through predictive models, data pipelines, and decision-focused dashboards. Hands-on across the full lifecycle from raw data handling and feature engineering to model deployment on cloud platforms like AWS and Azure. Recent work includes improving demand prediction, optimizing product-level decisions, and building systems that are actively used by business teams.

## Education

**Bennett University** — **Aug 2023 – May 2025**
*Master of Computer Application (MCA), Data Science (9.1 CGPA)* — *Greater Noida*

**J.C. Bose University of Science and Technology** — **Jul 2020 – Apr 2023**
*Bachelor of Computer Application (BCA), Data Science (8.5 CGPA)* — *Faridabad*

## Technical Skills

**Programming & Data Science**: Python, SQL, PySpark,C, Machine Learning, Deep Learning, Generative AI, Large Language Model (LLMs),Natural Langauge Processing (NLP), Data Modeling, Data Visualization, Extract Transform Load(ETL), Advanced MS Excel,ADLS,Postman.

**Libraries & Tools**: Agentic AI ,N8N,RAG,MCP,Docker,Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, SciPy, NLTK, Hugging Face, Beautiful Soup, SQLAlchemy, Power BI, Tableau, Jupyter, Google Colab, Git, Selenium,Linux,API

**Cloud Platforms**: AWS (SageMaker, EC2,S3, Lambda, Rekognition, Glue, Athena, Lake Formation, Bedrock), Azure (Databricks,Fabrics,Azure ML,Data Factory, Synapse, Blob Storage)

**DB**: SQL Developer, Postgresql, MYSQL, Oracle, MongoDB

**Soft Skills**: Teamwork, Communication, Leadership, Client Engagement, Presentation, Problem Solving

**Models**:Minimax,Anthropic,Open AI, Phi-2 / Phi-3 (Microsoft),OpenAI, Anthropic, Gemini (Google), Meta LLaMA 3, DeepSeek-VL / DeepSeek-Coder, Mistral 7B / Mixtral 8x7B,Random forest,XGBoost, and LightGBM.

## Experience

**Syren Cloud** (Databricks Partner)—HYD,IN — **Jul 2025 – Present**
*Data Scientist* — *Remote*

Building enterprise-scale ML systems across recommendation, MLOps automation, and analytics — on Microsoft Fabric and Databricks infrastructure.

- **Central Recommendation Engine (CRE):** Engineered core predictive algorithms processing multi-signal behavioral data for real-time personalization across globally supply chain markets.
- **MLOps UI Platform:** Architected ML backend abstracting model lifecycle (training, versioning, deployment) into a low-code interface for non-ML teams to operationalize Python models independently.
- **Hybrid ML Disaggregation Pipeline:** Designed a statistical system decomposing macro-level national KPIs into granular regional and SKU-level projections for precision supply planning.
- **Total Cooler Management System (TCMS):** Embedded lookalike models and autonomous KPI computation into production Power BI dashboards on Microsoft Fabric for self-refreshing field operations reporting.

**Appsquadz** (AWS Partner) — **Jan 2025 – Jun 2025**
*Data Scientist & AI Engineer* — *Noida*

Delivered end-to-end AI solutions across search, NLP, LLM fine-tuning, and analytics — deployed on AWS infrastructure for enterprise and consumer clients.

- **Semantic Video Search Engine:** Architected transcript-aware retrieval on AWS (Transcribe, OpenSearch, SageMaker, Bedrock) mapping natural-language queries to speaker-aware video timestamps.
- **Waves OTT Recommendation Engine:** Built a hybrid system fusing collaborative filtering with metadata embeddings (ANN) with cold-start fallback logic for a live streaming platform.
- **Mentor Policy Chatbot:** Fine-tuned LLaMA 3.1-8B-Instruct (LoRA/PEFT) on government policy corpus; deployed RAG pipeline via LangChain for scheme eligibility resolution at scale.
- **Enterprise Document Intelligence:** Deployed transformer-based NLP pipelines for content moderation, classification, and summarization; integrated Bhashini multilingual APIs and Cloud Vision OCR.
- **Soul Chat Automation:** Engineered adaptive analytics dashboards for a conversational AI platform surfacing session-level behavioral KPIs to product and operations stakeholders.

- **Churn & Sentiment Analytics:** Built ensemble churn prediction (85% accuracy) and real-time speech sentiment pipelines directly powering client CX retention workflows.

**Areness** **Jun 2024 – Dec 2024**
*Data Analyst* *Gurugram*

Owned end-to-end data pipeline and reporting infrastructure for legal compliance and operational analytics.

- **Scraping & Ingestion Pipelines:** Built fault-tolerant automated scraping (Selenium, BeautifulSoup) and ingestion framework (SQLAlchemy, Great Expectations) — reducing manual data intervention by 90%.
- **ETL & Data Quality:** Designed ETL pipelines with automated schema validation and quality gates — producing audit-ready datasets for legal compliance and regulatory KPI reporting.
- **Executive Dashboards:** Delivered Power BI dashboards giving senior stakeholders real-time visibility into legal operations, regulatory risk, and business performance metrics.

**S.O. Infotech Pvt. Ltd.** (Freelance) **Jul 2023 – May 2024**
*AI Engineer* *Hybrid*

Built and deployed NLP automation systems for legal document processing — covering full cycle from data preparation to production API integration.

- **Document Classification:** Trained BERT-based pipeline on preprocessed legal and insurance documents (PyMuPDF ingestion, regex cleaning) — achieving 92% accuracy across document categories.
- **Contract Entity Extraction:** Annotated training corpus and built domain-adapted SpaCy NER model extracting party names, clauses, and effective dates — reducing manual review time by 60%.
- **Model Evaluation & Deployment:** Iteratively improved precision/recall across ambiguous document classes; packaged both models as Flask REST APIs integrated into the client's intake management system.

**Golden Metal Corporation** **Jun 2022 – Jun 2023**
*AI Engineer* *Hybrid*

Delivered AI and analytics systems across forecasting, computer vision, and operational reporting for a manufacturing and procurement environment.

- **Demand Forecasting:** Benchmarked ARIMA and Prophet on historical procurement data to project raw material demand and seasonality — improving procurement efficiency by 20%.
- **Facial Recognition Attendance:** Built real-time face detection and recognition pipeline (OpenCV, TensorFlow) replacing manual tracking and strengthening facility access security.
- **Operational Dashboards:** Designed inventory optimization and supplier performance dashboards in Power BI and Streamlit — iterated directly with operations teams for visibility into material flow and vendor delays.
- **Report Automation:** Automated weekly reporting via Python scheduling with email triggers — eliminating ∼5 hours/week of manual effort across operations and management teams.

## Publications

**"Reactive to Agentic: Next Gen AI Agent Transition Framework"** **2025**
*IEEE 3rd Int'l Conf. on Advances in Computation, Communication and IT (ICAICCIT)*

**"Generative AI in Healthcare"** **2025**
*Book Chapter, ISBN: 9781032784847*

**"Leveraging Financial Data and Risk Management in the Banking Sector using Machine Learning"** **2024**
*IEEE BOMBAY 2023 9th International Conference for Convergence in Technology (I2CT)*

**"Comparative Analysis of Machine Learning Algorithms in Parkinson Disease Diagnosis"** **2024**
*IEEE 2023 2nd International Conference on Automation, Computing and Renewable Systems (ICACRS)*

## Certifications

- Python-AI (DUCAT, A+ Grade)
- C and C++ (Haryana State Council ITEC)
- Crash Course on Python (Google)
- Data Analysis with Python (IBM)
- Harnessing the Power of Data with Power BI (IBM)
- Data Analytics with Excel Pivot Tables (Microsoft)
- Extract, Transform, and Load Data in Power BI (Microsoft)
- Linux Fundamentals (LearnQuest)
- Data Mining Methods (University of Colorado Boulder)
- Data Analytics Methods for Marketing (META)